

Programming Fundamentals

Logic, Selections, and Iteration Control Structures

1.0 Selection Operators

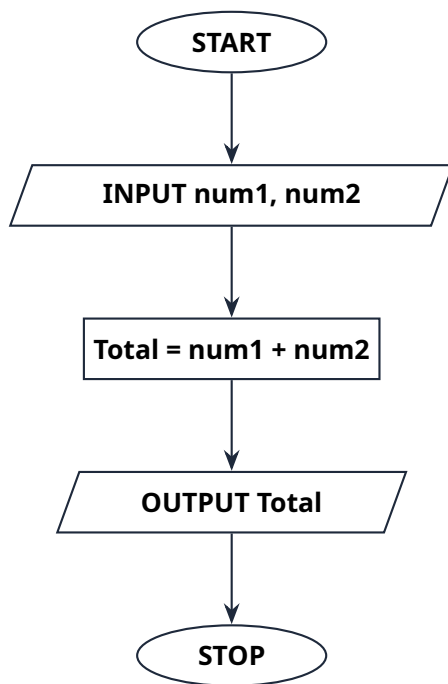
These operators are used within decision blocks (diamonds) to evaluate a condition as either True or False.

Operator	Description	Example Logic
<code>==</code>	Equality Check	<code>a == b</code> (Is a same as b?)
<code>!=</code> / <code><></code>	Inequality Check	<code>a != b</code> (Is a different?)
<code>></code> / <code><</code>	Magnitude	<code>a > b</code> (Is a strictly greater?)
<code>>=</code> / <code><=</code>	Inclusive Magnitude	<code>a >= b</code> (Is a greater/equal?)

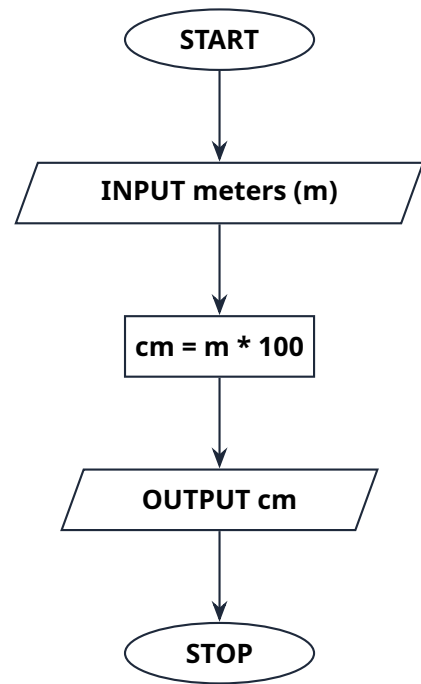
2.0 Sequential Logic Flowcharts

The following diagrams represent linear sequences where one task follows another without branching.

Addition Logic

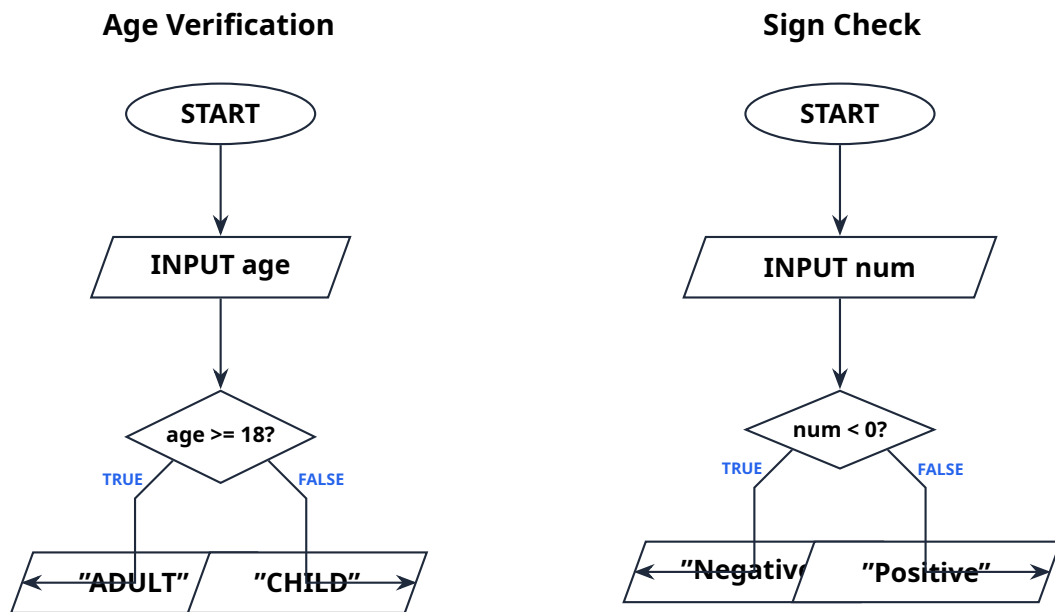


Unit Conversion (m to cm)



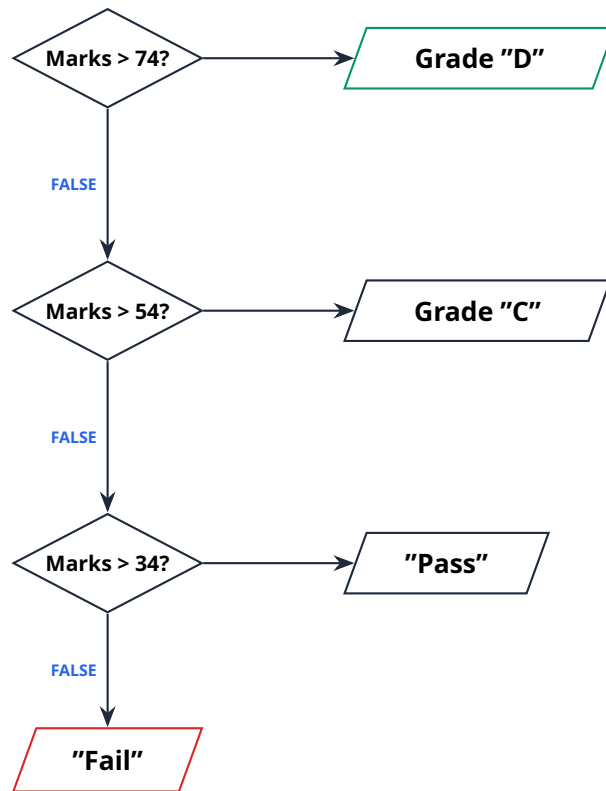
3.0 Simple Selection

Single decision point flowcharts for age verification and numerical sign checking.



4.0 Branching: Nested Grading System

This structure handles multiple outcomes by nesting decision diamonds.



5.0 Iteration Dry Run

Trace tables are used to manually execute code logic step-by-step to verify expected outputs.

Condition (count <= 6)	Display Result	Calculation	Final Count
$1 \leq 6$ (True)	1	$1 + 1$	2
$2 \leq 6$ (True)	2	$2 + 1$	3
$3 \leq 6$ (True)	3	$3 + 1$	4
$4 \leq 6$ (True)	4	$4 + 1$	5
$5 \leq 6$ (True)	PASS, 5	$5 + 1$	6
$6 \leq 6$ (True)	6	$6 + 1$	7
$7 \leq 6$ (False)	-	Exit Loop	7

6.0 Homework: Next Odd Logic

A decision flowchart to ensure the output is always the next consecutive odd number.

